MMDU MULLANA OFFICIAL SYLLABUS • BCSE-012 • UNIT 4

Unit 4: Cascading Style Sheets (CSS) Architecture (Cascade, Box Model & Layouts)

Subject: Fundamentals of Web Technologies (BCSE-012) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 40 mins

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1. Cascade, Specificity Weight Matrix & 3 Applying Methods

1. Introduction to Style Sheets & Specificity Calculation Formula

Cascading Style Sheets (CSS) describe the visual presentation and formatting of HTML documents, separating structural content from graphic design.

✅ **FUNDAMENTAL LAW / RULE:**
The CSS Specificity Weight Matrix Formula:

When conflicting rules target the same element, browsers calculate a 4-part weight:

$$\mathbf{Specificity} = (\text{Inline Style } [1000]) + (\text{IDs } [0100]) + (\text{Classes/Pseudo } [0010]) + (\text{Elements/Tags } [0001])$$

- `!important` overrides standard cascade weights and should be used with extreme caution.

- **Inheritance:** Typography properties (`color` , `font-family` , `font-size`) inherit by default to child elements; Box Model properties (`margin` , `padding` , `border` , `width`) do NOT inherit.

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2. Three Methods of Applying CSS

Method	Syntax Location	Scope	Browser Caching & Reusability
Inline CSS	Inside element tag:	Strictly that single individual element.	**Worst practice.** Pollutes HTML; cannot be cached independently.
Internal / Embedded	Inside		